State	Pseudocode	IR Ld	Reg Sel	Reg Wr	Reg En	A Ld	B Ld	ALUOp	ALU En	MA Ld	Mem Wr	Mem En	Imm Sel	Imm En	μBr	Next State
FETCH0:	MA ← PC; A ← PC	*	PC	0	1	1	*	*	0	1	0	0	*	0	N	*
	IR ← Mem	1	*	0	0	0	*	*	0	0	0	1	*	0	S	*
	PC ← A+4	0	PC	1	0	0	*	INC_A_4	1	*	0	0	*	0	D	*
. . .																
MAVX0:	microbranch back to FETCH0	*	*	0	0	*	*	*	0	*	0	0	*	0	J	FETCH0
	MA ← R[rs1]	0	rs1	0	1	*	*	COPY_A	0	1	0	0	*	0	N	
	A ← Mem	0	*	0	0	1	*	*	0	0	0	1	*	0	S	
	MA ← R[rs2]	0	rs2	0	1	*	*	COPY_B	0	1	0	0	*	0	N	
	B ← Mem	0	*	0	0	*	1	*	0	1	0	0	*	0	S	
LOOP	if (B < A) uBr to DONE	0	*	0	0	1	1	SLT	1	0	0	0	*	0	EZ	
	A ← A + 1	0	*	0	0	1	0	INC_A_1	1	*	0	0	*	0	N	
	uBr to LOOP	*	*	0	0	*	*	*	0	*	0	0	*	0	J	LOOP
DONE	MA ← R[rd]	*	rd	0	1	0	*	*	0	1	0	0	*	0	N	
	Mem ← A	*	*	0	0	0		CopyA	1	0	1	0	*	0	S	
	uBr to FETCH 0	*	*	0	0	*	*	*	0	*	0	0	*	0	J	FETCH0